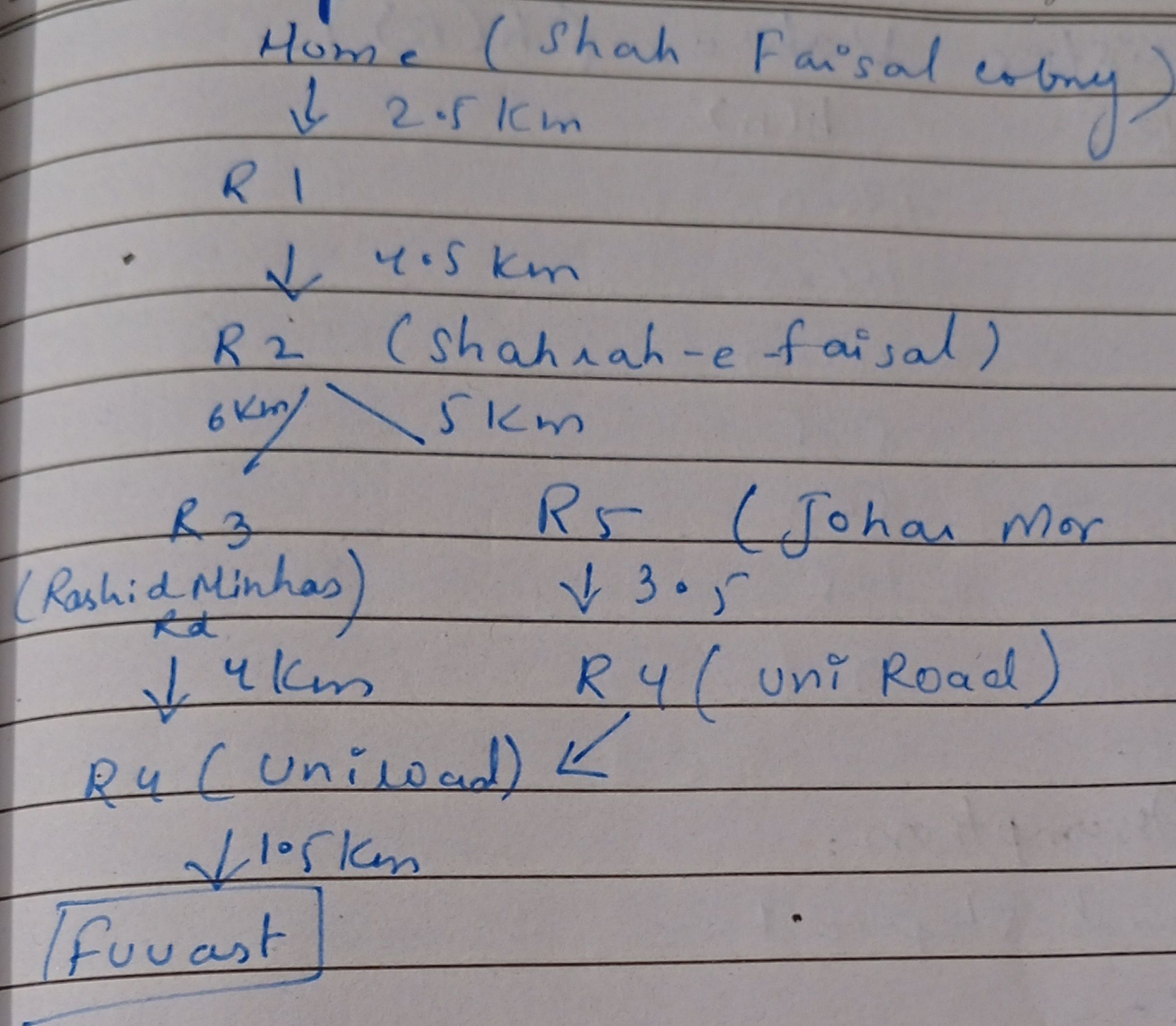


A* Search:-

Date _____
Map:-

Day _____



Connections:-

Home → R1

R1 → R2

R2 → R3, R5

R3 → R4

R5 → R4

R4 → Fuust

Date

Heuristic

Values (h(n))

Node
Map:-

$h(n)$

17.0

14.5

10.0

5.5

5.0

10.5

0

Forward

Cost Assumption:

Each road step = 1

Search:-

Step 0: Initialize

$PQ: [Home]$

$v: \emptyset$

$[g(Home) = 0]$

$[h(Home) = 5.2]$

$[f(Home) = g + h = 0 + 5.2 = 5.2]$

Parent Table:-

Home $\rightarrow$ None

Step 1: Expand Home:-

Neighbour: R1

Date

$PQ: [R1]$

$v: [Home]$

Node

R1

$[f(R1) =]$

choose

Parent

Home

R1 -

Step

Neigh

PQ

$v:$

No

R:

$f($

ch

PCQ: [R1]

V: [Home]

Node	g	h	f
R ₁	1	5.2	6.2

$$[f(R_1) = g + h = 1 + 5.2 = 6.2]$$

choose R₁ (f = 6.2)

Parent Table update:

Home → None

R₁ → Home

Step 2: Expand R₁

Neighbors: R₂

PCQ: [R₂]

V: [Home, R₁]

Node	g	h	f
R ₂	2	4.0	6.0

$$f(R_2) = 2 + 4.0 = 6.0$$

choose R₂ (f = 6.0)

Date _____

Parent Table update :

None $\rightarrow$ None

$R_1 \rightarrow$ Home

$R_2 \rightarrow R_1$

Step 3: Expand R_2

Neighbors: R_3, R_5

PQ: $[R_3, R_5]$

V: $[Home, R_1, R_2]$

For R_3 :

$[g = 3]$

$[h = 3 \cdot 3]$

$[f = 3 + 3 \cdot 3 = 6 \cdot 3]$

For R_5 :

$[g = 3]$

$[h = 4 \cdot 8]$

$[f = 3 + 4 \cdot 8 = 7 \cdot 8]$

Node

g

h

f

R_3

3

$3 \cdot 3$

$6 \cdot 3$

R_5

3

$4 \cdot 8$

$7 \cdot 8$

Enter text

Choose R_3 ($f = 6 \cdot 3$)

Parent Table

Home $\rightarrow$ None

$R_1 \rightarrow$ Home

$R_2 \rightarrow R_1$

$R_3 \rightarrow R_2$

$R_5 \rightarrow R_2$

Step 4: Expand

Neighbors

PQ: $[R_4]$

V: $[Home, R_1, R_2, R_3, R_5]$

For R_4 :

$[g = 4]$

$[h = 1 \cdot 0]$

$[f = 4 + 1 \cdot 0 = 4 \cdot 0]$

for R_5

$[g = 3]$

$[h = 4]$

$[f = 3 + 4 \cdot 8 = 7 \cdot 8]$

Node

R_4

R_5

Date _____
Parent Table update:-

Home $\rightarrow$ None

$R_1 \rightarrow$ Home

$R_2 \rightarrow R_1$

$R_3 \rightarrow R_2$

$R_5 \rightarrow R_2$

Step 4: Expand R_3

Neighbors: R_4

PQ: $[R_4, R_5]$

V: $[Home, R_1, R_2, R_3]$

for R_4 :

$[g = 4]$

$[h = 1.0]$

$[f = 4 + 1.0 = 5.0]$

for R_5 :

$[g = 3]$

$[h = 4.8]$

$[f = 3 + 4.8 = 7.8]$

Node	g	h	f
R_4	4	1.0	5.00
R_5	3	4.8	7.8

Date

Choose R_4 ($f=5.0$)

Parent Table update :-

Home $\rightarrow$ None

$R_1 \rightarrow$ Home

$R_2 \rightarrow R_1$

$R_3 \rightarrow R_2$

$R_5 \rightarrow R_2$

$R_4 \rightarrow R_3$

fu

Step 5: Expand R_4

Neighbour: Fuvast

PQ: [Fuvast, R_5]

V: [Home, R_1 , R_2 , R_3 , R_4]

for Fuvast

[$g=5$]

[$h=0$]

[$f=5+0=5.0$]

for R_5 :

[$g=3$]

[$h=4.8$]

[$f=3+4.8=7.8$]

Node

Fuvast

R_5

Choose Fuvast

Goal Found

Step 6: C

Final

Node

Home

R_1

R_2

R_3

R_5

R_4

Fuvast

Start

Fuvast

Rever

Home

Date _____

Day _____

Node	g	h	f
Fuvast	5	0	5=0
R5	3	4=8	7=8

Choose Fuvast ($f = 5 = 0$)

Goal found

Step 6: Goal found

Final Parent Table :

Node	Parent
Home	None
R1	Home
R2	R1
R3	R2
R5	R2
R4	R3
Fuvast	R4

Start from goal & go backward

Fuvast $\rightarrow$ R4 $\rightarrow$ R3 $\rightarrow$ R2 $\rightarrow$ R1 $\rightarrow$ Home

Reverse it:

Home $\rightarrow$ R1 $\rightarrow$ R2 $\rightarrow$ R3 $\rightarrow$ R4 $\rightarrow$ Fuvast